Encapsulation: Private fields ("-") + public accessors ("+") in model classes.
Inheritance (is-a): Shown with --|> arrows (User -> Student/CompanyRepresentative/Staff).
Composition: Repositories *-- domain objects (explicit multiplicity "1" to "0..*").
Aggregation: CompanyRepresentative o-- InternshipOpportunity indicates shared ownership.
Dependency Inversion: Services depend on repository abstractions (LoginService -> UserRepository).
Business Logic: InternshipService filters by year (Year 1-2: BASIC; Year 3-4: All) and major (case-insensitive).
Withdrawal Enhancement: Students can withdraw from both PENDING and SUCCESSFUL applications.
Self-Registration: Company representatives can register via LoginHandler (requires staff approval).
Filter System: OpportunityFilterService applies user-configured filters (Status, Major, Level, Date).
Filter Persistence: FilterStateManager maintains filter settings per user across menu navigation.
Default Behavior: All opportunities sorted alphabetically by title.

Model

«abstract»
User
-id: String
-name: String
-password: String
-role: UserRole
+getId(): String
+getName(): String
+setName(name: String): void
+getPassword(): String
+setPassword(pw: String): void
+getRole(): UserRole

UserRole
STUDENT
COMPANY_REPRESENTATIVE
CAREER_CENTER_STAFF

role
1
1

CareerCenterStaff
-department: String
+getDepartment(): String

CompanyRepresentative
-companyName: String
-department: String
-position: String
-approved: boolean
+isApproved(): boolean

ApplicationStatus
PENDING
SUCCESSFUL
UNSUCCESSFUL
WITHDRAWAL_REQUESTED

WithdrawalRequest
-id: String
-application: InternshipApplication
-reason: String
-decision: WithdrawalDecision
-requestedAt: LocalDateTime

WithdrawalDecision
PENDING
APPROVED
REJECTED

decision
creates
status
registrationRequests
withdrawalRequests

RegistrationRequest
-id: String
-representative: CompanyRepresentative
-approved: boolean
-reviewed: boolean
-submittedAt: LocalDateTime

InternshipOpportunity
-id: String
-title: String
-description: String
-level: InternshipLevel
-preferredMajor: String
-openingDate: LocalDate
-closingDate: LocalDate
-status: OpportunityStatus
-companyName: String
-createdBy: CompanyRepresentative
-maxSlots: int
-confirmedSlots: int
-visible: boolean
+isAvailable(): boolean

OpportunityStatus
PENDING
APPROVED
REJECTED
FILLED

InternshipLevel
BASIC
INTERMEDIATE
ADVANCED

Student
-yearOfStudy: int
-major: String
+getYearOfStudy(): int
+getMajor(): String

InternshipApplication
-id: String
-student: Student
-opportunity: InternshipOpportunity
-status: ApplicationStatus
-submittedAt: LocalDateTime
-placementAccepted: boolean
+isPlacementAccepted(): boolean

contains
student
applications
opportunity
level

returns

CLI

LoginHandler
-loginService: LoginService
-userRepository: UserRepository
-registrationRequestRepository: RegistrationRequestRepository
-idGenerator: IDGenerator
+login(): User
-registerCompanyRepresentative(): void

«abstract»
MenuBase
#userRepository: UserRepository
#internshipRepository: InternshipRepository
#applicationRepository: ApplicationRepository
#registrationRequestRepository: RegistrationRequestRepository
#withdrawalRequestRepository: WithdrawalRequestRepository
#internshipService: InternshipService
#opportunityFilterService: OpportunityFilterService
#filterStateManager: FilterStateManager
#currentUser: User
+display(): void
#handleChangePassword(): void
#configureFilters(): void
#clearFilters(): void

CompanyRepresentativeMenu
-representative: CompanyRepresentative
+display(): void

CareerCenterStaffMenu
+display(): void

StudentMenu
+display(): void
-viewAvailableInternships(): void
-applyForInternship(): void

CLIUtil
+readString(prompt: String): String
+readInt(prompt: String, min: int, max: int): int
+readDate(prompt: String): LocalDate
+displayInfo(msg: String): void

MainMenu
+display(): void

uses
creates requests
regRequestRepo
queries
userRepo
applicationRepo
authenticatesAgainst
delegates
creates
generates IDs
withdrawalRepo
internshipRepo
populates
launches

Repositories

RegistrationRequestRepository
-data: Map

UserRepository
-data: Map
+save(user: User): void
+findById(id: String): Optional
+findAll(): Collection

ApplicationRepository
-data: Map
+save(app: InternshipApplication): void
+findByStudentId(studentId: String): List

WithdrawalRequestRepository
-data: Map

InternshipRepository
-data: Map
+save(opp: InternshipOpportunity): void
+findById(id: String): Optional
+findAll(): Collection

contains
uses

Services

LoginService
-userRepository: UserRepository
+authenticate(id: String, password: String): User

internshipService

DefaultStudentApplicationService
-applicationRepository: ApplicationRepository
-withdrawalRequestRepository: WithdrawalRequestRepository
-internshipRepository: InternshipRepository
+getApplications(student: Student): List<InternshipApplication>
+getWithdrawableApplications(student: Student): List<InternshipApplication>
+submitApplication(student: Student, opp: InternshipOpportunity): InternshipApplication
+requestWithdrawal(app: InternshipApplication, reason: String): WithdrawalRequest
+acceptPlacement(app: InternshipApplication): void

filtersFrom
implements

«interface»
StudentApplicationService
+getApplications(student: Student): List<InternshipApplication>
+getPendingApplications(student: Student): List<InternshipApplication>
+getApplicationsAwaitingDecision(student: Student): List<InternshipApplication>
+getWithdrawableApplications(student: Student): List<InternshipApplication>
+findOpportunityById(id: String): Optional<InternshipOpportunity>
+hasExistingApplication(student: Student, opp: InternshipOpportunity): boolean
+submitApplication(student: Student, opp: InternshipOpportunity): InternshipApplication
+requestWithdrawal(app: InternshipApplication, reason: String): WithdrawalRequest
+acceptPlacement(app: InternshipApplication): void

InternshipService
-internshipRepository: InternshipRepository
+listAvailableInternships(): List<InternshipOpportunity>
+listAvailableInternshipsForStudent(student: Student): List<InternshipOpportunity>
-isEligibleForLevel(yearOfStudy: int, level: InternshipLevel): boolean
-isEligibleForMajor(studentMajor: String, preferredMajor: String): boolean

Config_Data_App

App
-config: AppConfig
+main(args: String[]): void

AppConfig
-userRepository: UserRepository
-internshipRepository: InternshipRepository
-applicationRepository: ApplicationRepository
-registrationRequestRepository: RegistrationRequestRepository
-withdrawalRequestRepository: WithdrawalRequestRepository
-dataBootstrap: DataBootstrap
-dataPersistenceManager: DataPersistenceManager
-idGenerator: IDGenerator
-reportGenerator: ReportGenerator
-internshipService: InternshipService
-studentApplicationService: StudentApplicationService
-opportunityFilterService: OpportunityFilterService
-filterStateManager: FilterStateManager
+getInternshipService(): InternshipService
+getStudentApplicationService(): StudentApplicationService
+getOpportunityFilterService(): OpportunityFilterService
+getFilterStateManager(): FilterStateManager

IDGenerator
+generateId(prefix: String): String

DataPersistenceManager
+exportAll(path: String): void

DataBootstrap
+initialize(): void

ReportGenerator
+generateReports(): void

initializes

Filter

FilterStateManager
-userFilterStates: Map<String, UserFilterState>
+getFilterState(userId: String): UserFilterState
+updateFilterCriteria(userId: String, criteria: OpportunityFilterCriteria): void
+clearFilters(userId: String): void

UserFilterState
-userId: String
-filterCriteria: OpportunityFilterCriteria
+getFilterCriteria(): OpportunityFilterCriteria
+setFilterCriteria(criteria: OpportunityFilterCriteria): void
+clearFilters(): void

OpportunityFilterCriteria
-status: OpportunityStatus
-preferredMajor: String
-level: InternshipLevel
-closingDateBefore: LocalDate
-sortAlphabetically: boolean
+hasActiveFilters(): boolean
+builder(): Builder

manages
uses
applies

OpportunityFilterService
+filterOpportunities(opportunities: List, criteria: OpportunityFilterCriteria): List
+matchesCriteria(opp: InternshipOpportunity, criteria: OpportunityFilterCriteria): boolean

filters